PHYSICS

Stage 6

Module 1: Kinematics

Motion on a Plane

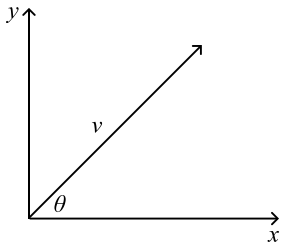
Teacher: Joshua Griffiths

Exam Equivalent Time: 60 minutes *(based on allocation of 1.5 minutes per mark)*

Questions

1. PHYSICS, M1 EQ-Bank 13

By separating the vector, shown below, into its constituent horizontal and vertical elements, express each component in terms of theta (θ). ***(2 marks)***



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2. PHYSICS, M1 EQ-Bank 3

A soccer ball leaves the ground with a velocity of 10 ms^{-1} at 40° above the horizontal.

i. Draw a vector diagram to represent the balls initial velocity. ***(1 marks)***

ii. Calculate, to 2 decimal place, the magnitude of the horizontal and vertical components of the vector. ***(2 marks)***

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3. PHYSICS, M1 EQ-Bank 17

A boy on a bike is travelling at 8 ms^{-1} north when he sees a girl on a scooter travelling at 6 ms^{-1} west, relative to the ground.

a. Draw a vector diagram to represent the velocity of the girl relative to the boy. ***(2 marks)***

b. Use the vector diagram to calculate the velocity of the girl relative to the boy. ***(2 marks)***

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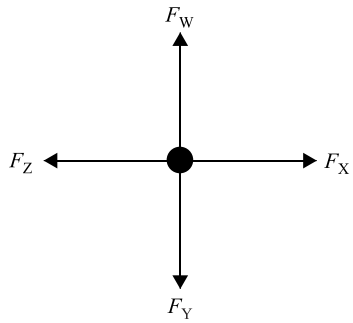
4. PHYSICS, M1 2018 VCE 5*

Four students are pulling on ropes in a four-person tug of war. The relative sizes of the forces acting on the various ropes are $F_W = 200\text{ N}, F_X = 240\text{ N}, F_Y = 180\text{ N}$ and $F_Z = 210\text{ N}$. The situation is shown in the diagram below.



What is the resultant force vector $(F_{\vec{R}})$ acting at the centre of the tug-of-war ropes? ***(3 marks)***

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5. PHYSICS, M1 EQ-Bank 14

A car is travelling north and approaching an intersection at 50 kmh^{-1}.

While maintaining a constant speed, the car turns left and continues east at 50 kmh^{-1}.

Using a vector diagram, calculate the change in velocity of the car. ***(3 marks)***

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6. PHYSICS, M1 EQ-Bank 15

Determine the magnitude and direction of the resultant velocity vector for an airplane that is simultaneously moving with a velocity of 350^{-1} directly towards the west and 275 ms^{-1} towards the northwest. ***(3 marks)***

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7. PHYSICS, M1 EQ-Bank 16

Car A is travelling north at 40 ms^{-1} and Car B is travelling at 30 ms^{-1} east.

Determine the velocity of Car A relative to Car B. ***(3 mark)***

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8. PHYSICS, M1 EQ-Bank 12

Below is a description of the motion of a runner. The motion can be divided into three stages.

Stage 1: Runner travels 120 metres south taking 20 seconds.

Stage 2: Runner turns west and travels at 5 ms^{-1} for half a minute.

Stage 3: Runner travels directly back to their starting position.

a. Determine the distance that the runner ran during Stage 2 of their journey. ***(1 mark)***

b. Find the displacement of the start point from the runner Stage 2 is completed. ***(3 marks)***

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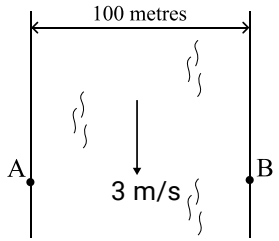
9. PHYSICS, M1 EQ-Bank 18

A physics student comes across a river which runs north to south and has a current of 3 ms^{-1} running south.



The student starts on the west side of the river at point A and paddles a kayak at 5 ms^{-1} directly across the river to finish at point B.

a. Calculate the angle which he must position the boat to travel in a straight line across the river. ***(2 mark)***

b. If the river is 100 metres wide, determine the time it takes for the student to cross the river. ***(2 mark)***

10. PHYSICS, M1 EQ-Bank 4

A plane that can fly at 500 kmh^{-1} with no wind, encounters a strong cross wind of 100 kmh^{-1} from the east. The plane needs to travel directly north to an airstrip

i. Calculate the angle, correct to one decimal place, at which the pilot should steer for the plane to fly directly to the airstrip? ***(3 marks)***

ii. If the airstrip is 100 km from the planes current position, how long, to the nearest minute, will it take for the plane to complete the journey? ***(2 marks)***

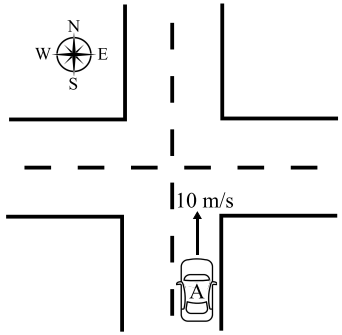
11. PHYSICS, M1 EQ-Bank 6

A boat is being rowed due north at a constant speed of 15 ms^{-1} when it encounters a current of 8 ms^{-1} going in the direction of S25°W.

Using vectors, determine the resultant velocity of the boat. ***(3 marks)***

12. PHYSICS, M1 EQ-Bank 5

Car A approaches an intersection at 10 ms^{-1} from the South as shown. As Car B approaches the intersection, it measures the velocity of Car A to be 22.36 ms^{-1} NW.



Draw Car B on the diagram showing its direction and speed. Show all working. ***(3 marks)***

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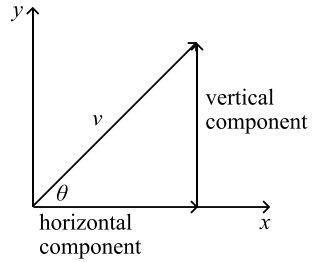
Worked Solutions

1. PHYSICS, M1 EQ-Bank 13

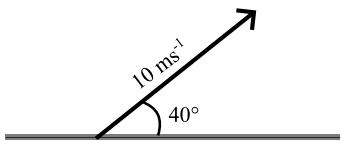


Horizontal component: $v\cos\theta$

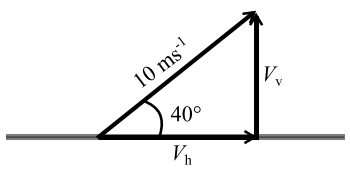
Vertical component: $v\sin\theta$

2. PHYSICS, M1 EQ-Bank 3

i.



ii.



$v_v = 10\sin 40° = 6.43\ \text{ms}^{-1}$

$v_h = 10\cos 40° = 7.66\ \text{ms}^{-1}$

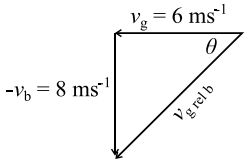
Worked Solutions

3. PHYSICS, M1 EQ-Bank 17

a. $v_g = 6\ \text{ms}^{-1}$ west

$-v_b = 8\ \text{ms}^{-1}$ south

$v_{g\,\text{rel}\,b} = v_g + (-v_b)$



b. $v_{g\,\text{rel}\,b} = \sqrt{6^2 + 8^2} = 10\ \text{ms}^{-1}$

$\tan\theta = \dfrac{8}{6}$

$\theta = \tan^{-1}\left(\dfrac{8}{6}\right) = 53.1°$

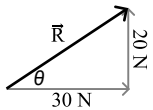
∴ Velocity of the girl relative to the boy is 10 ms^{-1}, S36.9°W.

4. PHYSICS, M1 2018 VCE 5*

→ By resolving the vertical vectors of $F_W = 200\ \text{N}$ up and $F_Y = 180\ \text{N}$ down, the net force in the vertical direction is $F_v = 20\ \text{N}$ up.

→ Similarly, the net force in the horizontal direction is $F_h = 30\ \text{N}$ to the right.

→ Thus the resultant vector $(F_{\vec{R}})$ can be calculated using the vector diagram below.



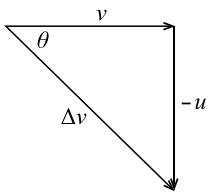
→ The magnitude of $F_{\vec{R}} = \sqrt{20^2 + 30^2}$ and $\theta = \tan^{-1}\left(\dfrac{20}{30}\right)$

→ Hence $F_{\vec{R}} = 36.1\ \text{N}, 33.7°$ above the horizontal.

5. PHYSICS, M1 EQ-Bank 14

$v = 50\ \mathrm{kmh}^{-1}$ east, $u = 50\ \mathrm{kmh}^{-1}$ north.

$\Delta v = v - u = v + (-u)$, where $-u = 50\ \mathrm{kmh}^{-1}$ south



$$\Delta v = \sqrt{50^2 + 50^2} = \sqrt{5000} = 70.7\ \mathrm{kmh}^{-1}$$

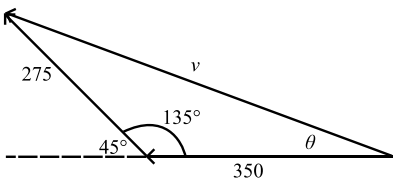
$$\tan\theta = \frac{50}{50}$$

$$\theta = \tan^{-1}\left(\frac{50}{50}\right) = 45°$$

Change in velocity of the car = 70.7 kmh^{-1}, S45°E.

6. PHYSICS, M1 EQ-Bank 15

Add vectors to show resultant vector:



Using the cosine rule to find the magnitude of v:

$$v = \sqrt{350^2 + 275^2 - 2 \times 350 \times 275 \times \cos 135°}$$
$$= 578.1376\ldots\ \mathrm{ms}^{-1}$$

Using the sign rule to find θ:

$$\frac{\sin\theta}{275} = \frac{\sin 135°}{578.1376}$$

$$\sin\theta = \frac{\sin 135° \times 275}{578.1376}$$

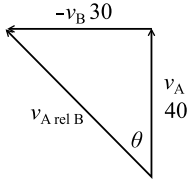
$$\theta = \sin^{-1}\left(\frac{\sin 135° \times 275}{578.1376}\right)$$
$$= 19.7°$$

∴ Resultant velocity of the aeroplane is 578 ms^{-1}, N70.3°W.

7. PHYSICS, M1 EQ-Bank 16

$v_{\text{A rel B}} = v_A - v_B = v_A + (-v_B)$

$v_A = 40\ \mathrm{ms}^{-1}$ north, $-v_B = 30\ \mathrm{ms}^{-1}$ west



$$v_{\text{A rel B}} = \sqrt{40^2 + 30^2} = 50\ \mathrm{ms}^{-1}$$

$$\tan\theta = \frac{30}{40}$$

$$\theta = \tan^{-1}\left(\frac{30}{40}\right) = 36.9°$$

Undefined control sequence \terefore Velocity of Car A relative to Car B is 50 ms^{-1}, N36.9°W.

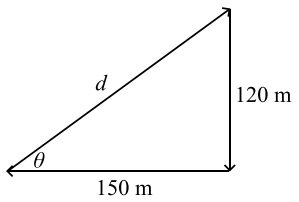
8. PHYSICS, M1 EQ-Bank 12

a. $d = v \times t$

$= 5 \times 30$

$= 150\ \text{m}$

b. Stage 1 and Stage 2 displacement diagram:



Using Pythagoras:

$d = \sqrt{150^2 + 120^2}$

$= \sqrt{36900}$

$= 192.1\ \text{m}$

$\tan\theta = \dfrac{120}{150}$

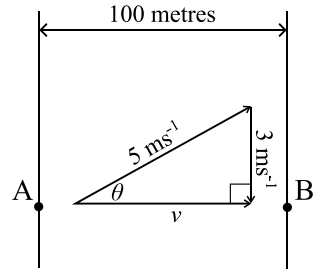
$\theta = \tan^{-1}\left(\dfrac{120}{150}\right)$

$= 38.7°$

∴ Displacement of the start point from the runner is 192.1 m, N51.3°E.

9. PHYSICS, M1 EQ-Bank 18

a.



$\sin\theta = \dfrac{3}{5}$

$\theta = \sin^{-1}\left(\dfrac{3}{5}\right) = 36.9°$

The student must turn 36.9° into the current as shown on the diagram.

b. Using Pythagoras:

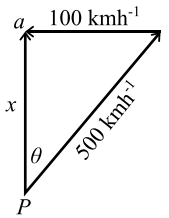
$v = \sqrt{5^2 - 3^2} = 4\ \text{ms}^{-1}$

$t = \dfrac{d}{s} = \dfrac{100}{4} = 25\ \text{s}$

∴ It will take the student 25 seconds to travel from A to B.

10. PHYSICS, M1 EQ-Bank 4

i.



$$\sin\theta = \frac{100}{500}$$

$$\theta = \sin^{-1}\left(\frac{100}{500}\right)$$

$$= 11.5°$$

Plane direction should be N11.5˚E

ii. Using the diagram in part (i):

$$\tan(11.5^\circ) = \frac{100}{x}$$

$$x = \frac{100}{\tan(11.5^\circ)}$$

$$= 491.5 \text{ kmh}^{-1}$$

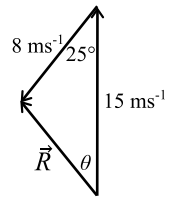
$$t = \frac{\text{distance}}{\text{speed}}$$

$$= \frac{100}{491.5}$$

$$= 0.203 \text{ h}$$

$$= 12 \text{ m (nearest minute)}$$

11. PHYSICS, M1 EQ-Bank 6



→ Using the cosine rule, the magnitude of $\vec{R}$ is:

$$\vec{R} = \sqrt{15^2 + 8^2 - 2 \times 15 \times 8 \times \cos 25^\circ}$$

$$= 8.455 \text{ ms}^{-1}$$

→ Using the sine rule, the direction of $\vec{R}$ is:

$$\frac{\sin\theta}{8} = \frac{\sin 25^\circ}{8.455}$$

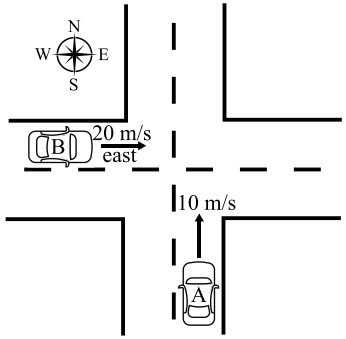
$$\sin\theta = \frac{8\sin 25^\circ}{8.455}$$

$$\theta = \sin^{-1}\left(\frac{8\sin 25^\circ}{8.455}\right)$$

$$= 23.6°$$

$$\therefore \vec{R} = 8.455 \text{ ms}^{-1} \text{ N23.6°W}$$

→ Velocity of A relative to B = $\boldsymbol{v_A - v_B = v_A + -v_B}$

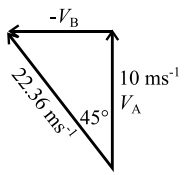


→ The direction of $\boldsymbol{-v_B}$ must be west, and therefore the direction of $\boldsymbol{v_B}$ is east.

→ Using Pythagoras theorem, the magnitude of $\boldsymbol{-v_B}$ is:

$$\begin{aligned} -v_B &= \sqrt{22.36^2 - 10^2} \\ &= 20\ \mathrm{ms}^{-1} \end{aligned}$$

→ Thus the vector $\boldsymbol{v_B}$ is $\mathbf{20\ ms^{-1}}$ east.



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